

JOSE MIGUEL DIEZ-CHAUPIS

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Objective Aspiring Quantitative Researcher concluding a rigorous academic tenure at UC Berkeley (Economics & CS). Currently tackling advanced coursework in Stochastic Processes (EECS 126) and Algorithmic Economics. Specialized in deep learning architectures (Mamba/SSM), econometric time-series analysis, and volatility modeling. Proven ability to synthesize economic theory with machine learning implementation in PyTorch.

EDUCATION

University of California, Berkeley | Berkeley, CA *Bachelor of Arts in Economics, Minors in Computer Science & Data Science* | Expected Aug 2026

Current Coursework (Spring 2026):

- **EECS 126: Probability and Random Processes:** Rigorous study of stochastic processes, Markov chains, Poisson processes, and estimation theory.
- **Econ 101A: Microeconomics (Quantitative):** Advanced math-intensive modeling of optimization, game theory, and market equilibrium.
- **Econ 143: Econometrics - Advanced Methods:** Applied analysis using instrumental variables, panel data, and time-series identification strategies.
- **Econ C147: Algorithmic Economics:** Intersection of CS and Economics, focusing on mechanism design, market microstructure, and computational complexity.

Foothill College | Los Altos Hills, CA *Associate of Science in Economics* | GPA: 3.91

TECHNICAL SKILLS

- **Languages & Frameworks:** Python, PyTorch, NumPy, Pandas, Scikit-learn, SQL.
 - **Quantitative Methods:** Stochastic Processes, Markov Chains, Ordinary Least Squares (OLS) Geometry, Vector Autoregression (VAR), State Space Models (SSM).
 - **Financial Concepts:** Volatility Modeling, Macro-Alpha Generation, Mechanism Design, Algorithmic Game Theory, Lead-Lag Analysis.
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QUANTITATIVE PROJECTS

Financial State Space Model (Mamba) Implementation | PyTorch, Deep Learning

<https://github.com/soccerpapo/pytorch-mamba-finance.git>

- Developed a custom "FinancialMambaBlock" architecture in PyTorch, adapting the Mamba State Space Model for high-frequency financial time-series forecasting.
- Engineered a specialized "physics" component within the model to constrain latent state dynamics to market realism.

- Implemented a custom decision head for trade signal generation, optimizing for inference speed and sequence modeling capabilities superior to traditional Transformers.
- Conducting ongoing testing on bias-variance trade-offs to ensure model robustness in non-stationary market regimes.

Copper-AI Macro Alpha Strategy | *Econometrics, Time-Series Analysis*

- Designed a Vector Autoregression (VAR) model to predict volatility in the Peruvian Sol (PEN) by analyzing semiconductor demand and global copper price fluctuations.
- Performed Lead-Lag analysis to identify predictive signals between industrial metal futures and FX spot rates.
- Backtested the strategy to isolate alpha generated from macro-economic structural dependencies rather than pure price momentum.

PROFESSIONAL EXPERIENCE

Logistics & Operations Manager (Independent Contractor) | Bay Area, CA 2020 – 2024

- Managed logistics and delivery operations for high-volume platforms, utilizing route optimization algorithms (mental heuristics) to maximize revenue per hour.
- Demonstrated disciplined time management and operational execution under tight constraints.

Junior Quantitative Estimator | DoorSpot Co. LLC 2019 – 2020

- Assisted in cost estimation and quantitative analysis for project bids.
- Managed data inputs for estimation models, ensuring accuracy in material and labor projections.

LEADERSHIP & AFFILIATIONS

Collegiate Athletics

- **Varsity Soccer Athlete:** Balanced rigorous academic schedule with high-performance athletic training, developing discipline, teamwork, and resilience under pressure.